Superpixel-Based Bipartite Graph Clustering Enriched With Spatial Information for Hyperspectral and LiDAR Data

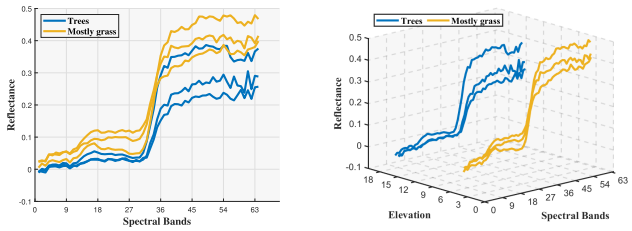
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***Abstract*— The surge in remote sensing (RS) data underscores the need for improved data diversity and processing. While integrating hyperspectral (HS) and light detection and ranging (LiDAR) data enhances analysis and addresses spectral variability, the high dimensionality, noise, and outliers inherent in hyperspectral images present significant challenges. In addition, the precise labeling required for HS makes supervised classification labor-intensive, professional-focused, and time-consuming, further motivating the development of advanced HS clustering algorithms to address these issues. Unsupervised clustering addresses the above issues but still struggles due to the underutilization of auxiliary spatial and structural information, high data dimensionality with redundant hyperspectral bands, and information divergence from heterogeneity among multimodal data. These challenges impede the effective extraction of consistent structures, undermining clustering stability and overall model performance. To address these challenges, we propose a superpixel-based bipartite graph clustering (SBGC) enriched with spatial information for hyperspectral and LiDAR data models. Our proposed method fully utilizes spatial information to construct meaningful bipartite graphs for the efficient processing of multimodal RS data. By adopting a projected clustering paradigm, our approach simultaneously clusters and reduces dimensionality, effectively eliminating redundant bands. In addition, it innovatively stacks multimodal data into tensors, thoroughly exploring the consistent structures in the low-rank space among different modalities. This reduces the heterogeneity-induced information divergence and significantly enhances clustering performance. Extensive experiments on real datasets confirm the method's effectiveness and advanced capabilities.**

***Index Terms*— Bipartite graphs, dimensionality reduction (DR), remote sensing (RS), scalable method, spatial information, tensor-based clustering, unsupervised learning.**

I. Introduction

EARTH observation community has recently seen a significant increase in remote sensing (RS) data [1], [2], emphasizing the need to enhance data diversity and capabilities [3]. This advancement is crucial for addressing the complex challenges and growing demands of modern RS applications [4]. Single-modality data often lack comprehensive detail, but integrating multiple data types provides a richer, more nuanced understanding [5]. For instance, hyperspectral (HS) images capture reflectance across a wide spectrum of wavelengths, offering high spectral resolution over large spatial areas, and can be collected through various platforms such as drones or orbital spectrometers. These detailed spectral signatures are invaluable for applications such as land use and cover identification or spectral unmixing. Multimodal data, such as the combination of HS and light detection and ranging (LiDAR) data, provide complementary insights that mitigate the limitations of single-modality observations [6]. While HS data offer material-specific information, LiDAR provides precise elevation measurements, and synthetic aperture radar (SAR) [7] adds insights into surface roughness and moisture content. Combining these diverse data sources gives a more comprehensive view of regions of interest. For instance, combining SAR and optical data enhances surface characterization, and recently, the integration of HS and LiDAR data has significantly addressed spectral variability issues associated with HS data. As illustrated in Fig. 1, their joint use effectively alleviates this problem. Consequently, joint land cover recognition using multimodal RS data has become a promising research direction in both the RS and machine learning communities, garnering widespread attention and becoming a leading research hotspot.



(a) (b)

Fig. 1. Spectral band curves and elevation data for two easily confused land cover types from the MUUFL dataset. While Trees and Mostly grass show overlapping spectral curves, their elevation distributions enable clear differentiation. This highlights the importance of integrating HS and LiDAR data. (a) Spectral bands of land cover. (b) Elevation of land cover.

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Land-cover material recognition using multimodal RS data can be divided into two types of tasks: classification [8], [9], [10] and clustering [11], [12] based on the learning scheme used. Over the past few decades, supervised classification methods for multimodal RS data [13], [14] have evolved from shallow to deep models, developing models with labeled training pixels to assign specific labels to each testing pixel. Notable advances have been made by Kong et al. [15], Han et al. [16], and Wang et al. [17] in the joint classification of HS and LiDAR data. These methods focus on integrating features from original images to form compact and discriminative representations, relying on advanced feature extractors and fusion strategies. With the continuous advancement of deep learning models such as CNNs and RNNs, the performance of feature-level fusion models has significantly improved [18], [19], [20]. In addition, transformer models [21], graph neural networks (GNNs) [22], and graph convolution networks (GCNs) [23] have enhanced the capture of complex spatial and spectral dependencies. Decision-level fusion approaches typically use adaptive weighting schemes and ensemble learning techniques to merge predictions from different data sources, yielding joint classification results [24]. Information interaction approaches enhance feature discrimination by incorporating clues from one modality into another's feature extraction process. Strategies such as weight sharing, cross-attention, and contrastive learning exploit interdependent knowledge from heterogeneous features, leading to excellent results. Recent advances include multiscale feature aggregation and hybrid attention mechanisms, further improving classification accuracy and robustness [25]. All in all, supervised classification methods for multimodal RS have shown remarkable success in land-cover material recognition for Earth observation.

However, the performance of the supervised learning scheme strongly relies on annotating HS data, which is labor-intensive, professional-focused, and time-consuming. The labeling of HS is particularly challenging, as it requires specialized domain knowledge and often involves extensive fieldwork for accurate expert annotations, making it a costly and resource-intensive task [26]. HS analysis without label guidance is further complicated by large spectral variability and complex spatial structures. Thus, developing an effective HS and LiDAR classification model remains challenging due to the need for extensive labeled pixels to achieve high accuracy, which is both expensive and limited. This motivates the exploration of unsupervised clustering algorithms as an alternative for analyzing HS data, enabling meaningful insights without the heavy reliance on labeled pixels.

In contrast, unsupervised joint clustering of multimodal RS data provides a method to circumvent the need for human-annotated labels by utilizing unlabeled pixels and partitioning them into clusters without prior knowledge. Despite its potential, this approach has been relatively underexplored, as indicated by the limited research conducted by Shahi et al. [27]. The primary reason for the scarcity of studies in unsupervised multimodal RS learning lies in the inherent variability and complexity of multimodal RS objects [28]. This complexity hinders the fusion of complementary information and semantic alignment without reference data and makes measuring similarity between pixels in heterogeneous RS data challenging. To address these challenges, numerous graph-based methods have been proposed to explore relationships between pixels using weighted graphs and conducting partitions based on similarities [29], [30], [31]. However, these methods often overlook auxiliary spatial and structural information and require significant computational and memory resources, especially for large-scale data due to their reliance on the affinity matrix and Eigen decomposition.

Consequently, unsupervised clustering of multimodal RS data remains a challenging task. In addition to the previously mentioned issues of ignoring auxiliary spatial and structural information and high computational complexity, particularly with large-scale data, hyperspectral data contain many redundant bands due to the high correlation between sequential feature bands [32], [33]. This redundancy results in high data dimensionality and processing complexity [34]. Furthermore, the heterogeneity among multimodal data can lead to information divergence, preventing models from effectively extracting the consistent structures within each modality [35], [36], [37]. This undermines the initial goal of multimodal data to provide complementary insights that mitigate the limitations of single-modality observations, leading to unstable clustering results and reducing the effectiveness of the models.

Based on the aforementioned issues, we propose a superpixel-based bipartite graph clustering (SBGC) enriched with spatial information for hyperspectral and LiDAR data. The main contributions of our algorithm are given as follows.

1) To address the underutilization of spatial information and high time complexity, we use the entropy rate superpixel (ERS) segmentation [38] method to construct a bipartite graph, effectively leveraging high-resolution RS data to establish fewer yet more representative anchors. Incorporating spatial information into the model is crucial for improving clustering quality, as pixels in close proximity often belong to the same class and exhibit similar spectral characteristics. This spatial regularization helps mitigate the effects of noise and spectral variability commonly found in hyperspectral (HS) images. By embedding spatial context into the clustering algorithm, the approach not only reduces time and computational costs but also enhances performance and ensures high-quality partitions in HS analysis.
2) To tackle the issue of high data dimensionality and complexity caused by redundant bands in hyperspectral data, our method employs a projection clustering paradigm that integrates dimensionality reduction (DR) and clustering. This approach effectively removes redundant information in hyperspectral data while preserving the most significant ones, thereby simplifying data processing and enhancing overall performance.
3) To address the challenge of information divergence arising from the heterogeneity among multimodal data, we innovatively stack multimodal data into tensors, thoroughly exploring the consistent structures in the low-rank space across different modalities. This approach reduces information divergence caused by

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heterogeneity and effectively integrates the rich complementary information from various modalities, thereby ensuring stability and enhancing clustering performance.

II. Notations and Preliminaries

In this article, we denote third-order tensors using bold calligraphic letters, e.g., $\boldsymbol{\mathcal{A}}$, matrices with bold uppercase letters, e.g., $\boldsymbol{A}$, vectors with bold lowercase letters, e.g., $\boldsymbol{a}$, and scalars with lowercase letters, e.g., a. The field of real numbers is denoted by $\mathbb{R}$. For a matrix $\boldsymbol{A} \in \mathbb{R}^{n_1 \times n_2}$, $\boldsymbol{A}^T$ denotes its transpose and $\mathtt{tr}(\boldsymbol{A})$ denotes its trace. The identity matrix of size $N \times N$ is represented by $\boldsymbol{I} \in \mathbb{R}^{N \times N}$ and the matrix of ones by $\boldsymbol{1} \in \mathbb{R}^{N \times 1}$.

For a third-order tensor $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its (i, j, k)th entry is denoted by $\boldsymbol{\mathcal{A}}_{ijk}$ and the tube by $\boldsymbol{\mathcal{A}}(i, j, :)$. Using MATLAB notation, $\boldsymbol{\mathcal{A}}(k, :, :)$, $\boldsymbol{\mathcal{A}}(:, k, :)$, and $\boldsymbol{\mathcal{A}}(:, :, k)$ denote the kth horizontal, lateral, and frontal slices, respectively [39]. The kth frontal slice, $\boldsymbol{\mathcal{A}}(:, :, k)$, can also be succinctly represented as $\boldsymbol{A}^k$. In addition, $\tilde{\boldsymbol{A}} = \mathtt{fft}(\boldsymbol{\mathcal{A}}, [], 3)$ denotes the fast Fourier transform (FFT) [40] applied along the third dimension of $\boldsymbol{\mathcal{A}}$. The inverse FFT (IFFT) [40] is used to revert $\tilde{\boldsymbol{A}}$ to $\boldsymbol{\mathcal{A}}$, expressed as $\boldsymbol{\mathcal{A}} = \mathtt{ifft}(\tilde{\boldsymbol{A}}, [], 3)$.

A. Discrete Fourier Transform

The discrete Fourier transform (DFT) plays a central role in the T-product discussed later. We provide some background and notation here. The DFT of a vector $\boldsymbol{v} \in \mathbb{R}^n$, denoted as $\tilde{\boldsymbol{v}}$, is given by

$$\tilde{\boldsymbol{v}} = \boldsymbol{F}_n \boldsymbol{v}, \quad \tilde{\boldsymbol{v}} \in \mathbb{C}^n \tag{1}$$

where $\boldsymbol{F}_n \in \mathbb{C}^{n \times n}$ is the DFT matrix defined as

$$\boldsymbol{F}_n = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \cdots & \omega^{(n-1)(n-1)} \end{bmatrix} \tag{2}$$

where $\omega = e^{-(2\pi i/n)}$ is a primitive nth root of unity with $i = \sqrt{-1}$. Note that $\boldsymbol{F}_n/\sqrt{n}$ is a unitary matrix, i.e.,

$$\boldsymbol{F}_n^* \boldsymbol{F}_n = \boldsymbol{F}_n \boldsymbol{F}_n^* = n\boldsymbol{I}_n. \tag{3}$$

Now, we consider the DFT applied to tensors. For $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, we denote $\tilde{\boldsymbol{A}} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$ as the result of applying the DFT to $\boldsymbol{\mathcal{A}}$ along the third dimension, i.e., performing the DFT on all the frontal slices of $\boldsymbol{\mathcal{A}}$. Using the MATLAB command `fft`, we have

$$\tilde{\boldsymbol{A}} = \mathtt{fft}(\boldsymbol{\mathcal{A}}, [], 3). \tag{4}$$

Similarly, we can recover $\boldsymbol{\mathcal{A}}$ from $\tilde{\boldsymbol{A}}$ using the IFFT, i.e.,

$$\boldsymbol{\mathcal{A}} = \mathtt{ifft}(\tilde{\boldsymbol{A}}, [], 3). \tag{5}$$

B. T-Product and T-SVD

Definition 1 (T-Product [41]): Let $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\boldsymbol{\mathcal{B}} \in \mathbb{R}^{n_2 \times l \times n_3}$. The T-product $\boldsymbol{\mathcal{A}} * \boldsymbol{\mathcal{B}}$ is defined as a tensor $\boldsymbol{\mathcal{C}}$ of size $n_1 \times l \times n_3$ given by

$$\boldsymbol{\mathcal{C}} = \boldsymbol{\mathcal{A}} * \boldsymbol{\mathcal{B}} = \mathtt{fold}(\mathtt{bcirc}(\boldsymbol{\mathcal{A}}) \cdot \mathtt{unfold}(\boldsymbol{\mathcal{B}})) \tag{6}$$

where

$$\mathtt{bcirc}(\boldsymbol{\mathcal{A}}) = \begin{bmatrix} \boldsymbol{A}^1 & \boldsymbol{A}^{n_3} & \cdots & \boldsymbol{A}^2 \\ \boldsymbol{A}^2 & \boldsymbol{A}^1 & \cdots & \boldsymbol{A}^3 \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A}^{n_3} & \boldsymbol{A}^{n_3-1} & \cdots & \boldsymbol{A}^1 \end{bmatrix} \tag{7}$$

$$\mathtt{unfold}(\boldsymbol{\mathcal{A}}) = \begin{bmatrix} \boldsymbol{A}^1 \\ \boldsymbol{A}^2 \\ \vdots \\ \boldsymbol{A}^{n_3} \end{bmatrix}, \quad \mathtt{fold}(\mathtt{unfold}(\boldsymbol{\mathcal{A}})) = \boldsymbol{\mathcal{A}}. \tag{8}$$

The T-product is analogous to matrix multiplication in the Fourier domain. Specifically, $\boldsymbol{\mathcal{C}} = \boldsymbol{\mathcal{A}} * \boldsymbol{\mathcal{B}}$ can be viewed as $\tilde{\boldsymbol{C}} = \tilde{\boldsymbol{A}}\tilde{\boldsymbol{B}}$. This property suggests that the FFT-based approach outlined in Algorithm 1 can be used to compute the T-product instead of directly applying (6).

Algorithm 1 T-Product

Input: $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\boldsymbol{\mathcal{B}} \in \mathbb{R}^{n_2 \times l \times n_3}$.
Output: $\boldsymbol{\mathcal{C}} = \boldsymbol{\mathcal{A}} * \boldsymbol{\mathcal{B}} \in \mathbb{R}^{n_1 \times l \times n_3}$.

1. Compute $\tilde{\boldsymbol{A}} = \mathtt{fft}(\boldsymbol{\mathcal{A}}, [], 3)$ and $\tilde{\boldsymbol{B}} = \mathtt{fft}(\boldsymbol{\mathcal{B}}, [], 3)$.
2. Compute each frontal slice of $\tilde{\boldsymbol{C}}$ by

$$\tilde{\boldsymbol{C}}^i = \begin{cases} \tilde{\boldsymbol{A}}^i \tilde{\boldsymbol{B}}^i, & i = 1, \ldots, \lceil \frac{(n_3+1)}{2} \rceil, \\ \mathtt{conj}(\tilde{\boldsymbol{C}}^{n_3-i+2}), & i = \lceil \frac{n_3+1}{2} \rceil + 1, \ldots, n_3. \end{cases}$$

3. Compute $\boldsymbol{\mathcal{C}} = \mathtt{ifft}(\tilde{\boldsymbol{C}}, [], 3)$.

Definition 2 (f-Diagonal Tensor [41]): A tensor is referred to as a frontal diagonal if each of its frontal slices consistently manifests as a diagonal matrix.

Definition 3 (Tensor Transpose [41]): The transpose of a tensor $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is the tensor $\boldsymbol{\mathcal{A}}^T \in \mathbb{R}^{n_2 \times n_1 \times n_3}$ obtained by transposing each of the frontal slices and then reversing the order of transposed frontal slices 2 through n_3.

As an example, let $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times 4}$ and its frontal slices be $\boldsymbol{A}^1$, $\boldsymbol{A}^2$, $\boldsymbol{A}^3$, and $\boldsymbol{A}^4$. Then,

$$\boldsymbol{\mathcal{A}}^T = \mathtt{fold}\left(\begin{bmatrix} \boldsymbol{A}^{1T} \\ \boldsymbol{A}^{4T} \\ \boldsymbol{A}^{3T} \\ \boldsymbol{A}^{2T} \end{bmatrix}\right). \tag{9}$$

Definition 4 (T-SVD [42]): Given $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, it is possible to factorize it as follows:

$$\boldsymbol{\mathcal{A}} = \boldsymbol{\mathcal{U}} * \boldsymbol{\mathcal{S}} * \boldsymbol{\mathcal{V}}^T \tag{10}$$

where $\boldsymbol{\mathcal{U}} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\boldsymbol{\mathcal{V}} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors, $\boldsymbol{\mathcal{S}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is a tensor with an f-diagonal structure, and $\boldsymbol{\mathcal{V}}^T$ is the transpose of $\boldsymbol{\mathcal{V}}$. The details of the decomposition process can be found in Algorithm 2.

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Algorithm 2 T-SVD

Input: $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$.

Output: T-SVD components $\mathcal{U}$, $\mathcal{S}$ and $\mathcal{V}$ of $\mathcal{A}$.

1. Compute $\tilde{\boldsymbol{A}} = \texttt{fft}(\mathcal{A}, [], 3)$.
2. Compute each frontal slice of $\tilde{\boldsymbol{U}}$, $\tilde{\boldsymbol{S}}$ and $\tilde{\boldsymbol{V}}$ from $\tilde{\boldsymbol{A}}$ by

for $i = 1, \ldots, \lceil \frac{n_3+1}{2} \rceil$ **do**
$[\tilde{\boldsymbol{U}}^i, \tilde{\boldsymbol{S}}^i, \tilde{\boldsymbol{V}}^i] = \text{SVD}(\tilde{\boldsymbol{A}}^i)$;
end for
for $i = \lceil \frac{n_3+1}{2} \rceil + 1, \ldots, n_3$ **do**
$\tilde{\boldsymbol{U}}^i = \texttt{conj}(\boldsymbol{U}^{n_3-i+2})$;
$\tilde{\boldsymbol{S}}^i = \tilde{\boldsymbol{S}}^{n_3-i+2}$;
$\tilde{\boldsymbol{V}}^i = \texttt{conj}(\tilde{\boldsymbol{V}}^{n_3-i+2})$;
end for

3. Compute $\mathcal{U} = \texttt{ifft}(\tilde{\boldsymbol{U}}, [], 3)$, $\mathcal{S} = \texttt{ifft}(\tilde{\boldsymbol{S}}, [], 3)$ and $\mathcal{V} = \texttt{ifft}(\tilde{\boldsymbol{V}}, [], 3)$.

Definition 5 (Tensor Nuclear Norm [42]): The nuclear norm of a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, denoted as $\|\mathcal{A}\|_*$, is defined as follows:

$$\|\mathcal{A}\|_* = \frac{1}{n_3}\|\tilde{\boldsymbol{A}}\|_* = \frac{1}{n_3}\sum_{i=1}^{n_3}\|\tilde{\boldsymbol{A}}^i\|_* = \frac{1}{n_3}\sum_{i=1}^{n_3}\sum_{j=1}^{l}\sigma_j(\tilde{\boldsymbol{A}}^i) \tag{11}$$

where $\tilde{\boldsymbol{A}}$ is the Fourier domain representation of $\mathcal{A}$, $\tilde{\boldsymbol{A}}^i$ denotes the ith frontal slice of $\tilde{\boldsymbol{A}}$, $l = \min(\text{n}_1, \text{n}_2)$, and $\sigma_j(\tilde{\boldsymbol{A}}^i)$ represents the singular values of $\tilde{\boldsymbol{A}}^i$.

III. Proposed Method

A. Overview

Clustering is an unsupervised learning technique that groups similar objects or data points without the need for GT labels or expert annotations. We denote the pixels in a hyperspectral (HS) image as a set $\mathbf{X}^1 = \{x_i^1\}_{i=1}^N \in \mathbb{R}^{B^1}$ and the pixels in a LiDAR image as a set $\mathbf{X}^2 = \{x_i^2\}_{i=1}^N \in \mathbb{R}^{B^2}$, where x_i^1 and x_i^2 represent the spectral signatures of a pixel, B^1 and B^2 are the respective numbers of spectral bands, and N is the total number of pixels. Clustering algorithms partition pixels into a clustering $\{x_k\}_{k=1}^c$ (where each x_k is a cluster and c is the number of clusters) such that pixels from the same cluster are "similar" (possibly due to a similar material constitution), while pixels from different clusters are "dissimilar" (containing different materials). The specific notion of similarity used varies widely across the many clustering algorithms in the literature.

To understand the key advancements of the proposed SBGC enriched with spatial information for hyperspectral and LiDAR data, it is essential to explore its four pivotal aspects: spatiality, deredundancy, efficiency/scalability, and consistency. Each of these components addresses critical challenges in RS data processing, ensuring improved performance and practicality.

1) *Spatiality:* SBGC employs the ERS segmentation approach to build a bipartite graph, effectively integrating spatial context with spectral information to address spectral variability. By incorporating spatial details from high-resolution RS data, SBGC is able to use contextual information from nearby pixels, which often belong to the same class and share similar spectral properties. This spatial integration is crucial for mitigating noise and improving the quality of clustering results, as spatially proximate pixels typically exhibit homogeneous characteristics. The ERS segmentation method generates compact, homogeneous superpixels by minimizing graph cut costs, which enhances the algorithm's ability to capture spatial structure and boundary delineation, as demonstrated in segmentation benchmarks. Compared to traditional methods, SBGC selects fewer but more representative anchors, optimizing both time and computational resources. This focus on a smaller, highly informative set of anchors ensures that the spatial and textural information is effectively utilized, maintaining model performance while reducing computational complexity.
2) *Deredundancy:* SBGC integrates a projection clustering paradigm. This approach combines DR with clustering techniques. It efficiently eliminates redundant information while preserving the most critical features of the hyperspectral data. As a result, the data processing becomes more straightforward, and the overall performance of the model is enhanced. This reduction in dimensionality simplifies data handling and improves the model's ability to process and analyze hyperspectral data effectively.
3) *Efficiency/Scalability:* SBGC utilizes the ERS segmentation approach not only to address the underutilization of spatial information but also to tackle the high time complexity associated with processing large-scale RS data. By constructing a bipartite graph with fewer but more representative anchors, the method significantly reduces computational costs and processing time. This approach ensures that the system remains efficient while maintaining high performance, making it more practical for large-scale applications in RS where computational resources and processing speed are critical factors.
4) *Consistency:* SBGC innovatively stacks multimodal data into tensors. This approach allows for a thorough exploration of consistent structures within a low-rank space across different modalities. By doing so, the method effectively integrates the complementary information from various data sources, reducing the divergence and ensuring that the rich information from each modality is utilized cohesively. This innovation enhances the stability and performance of clustering, making it possible to achieve more accurate and reliable results in multimodal RS applications, overcoming the issue of information divergence caused by the heterogeneity among multimodal data.

B. Integrating Spatial Information for Anchor Selection

Traditional graph-based clustering methods fail to adequately utilize spatial information in hyperspectral data and cannot handle large-scale data. To address this, we introduce

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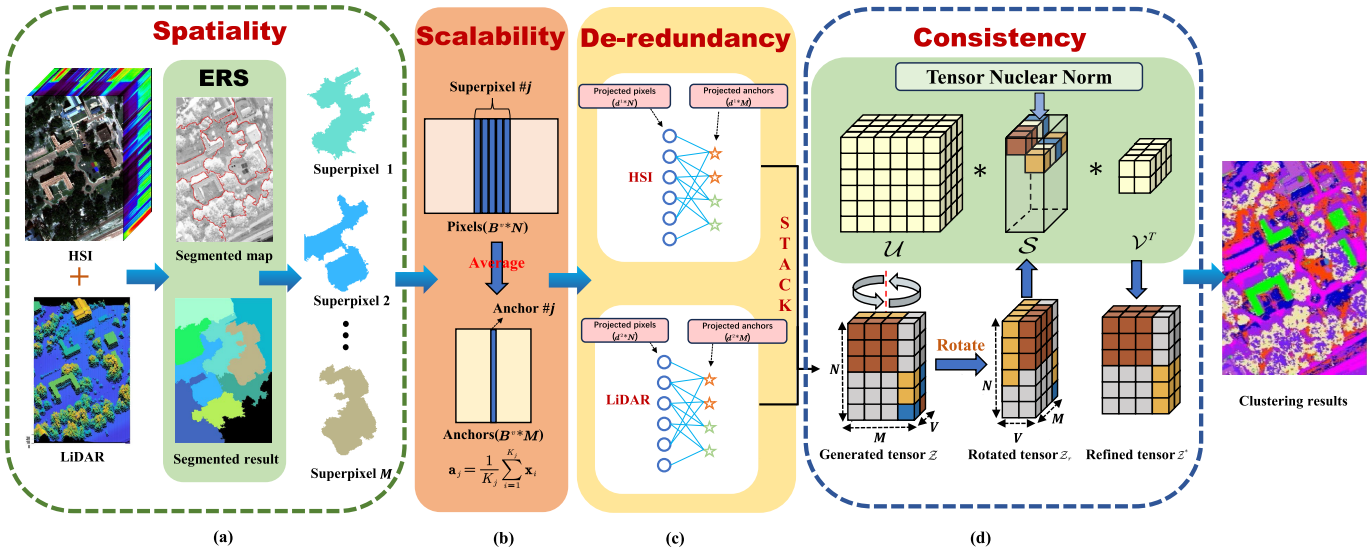


Fig. 2. SBGC method for HS and LiDAR data improves performance by focusing on four key aspects: Spatiality, employing the ERS segmentation algorithm to generate compact and homogeneous superpixels; deredundancy, combining DR and clustering; efficiency/scalability, using fewer representative anchors to lower costs; and consistency, integrating multimodal RS data into tensors for stable, accurate cluster results. (a) Generate superpixels. (b) Generate anchors. (c) Construct bipartite graphs. (d) Extracting consistent information.

a bipartite graph-based clustering method (anchor graph) that significantly reduces computational complexity. Recognizing the limitations of traditional anchor selection methods, which do not fully leverage spatial information in hyperspectral data, we innovatively combine the ERS segmentation method to propose a new anchor selection strategy. This strategy better utilizes the spatial information, selecting fewer but more representative anchors, resulting in a low-dimensional bipartite graph. This approach significantly reduces time and space complexity, especially when processing large-scale datasets.

1) Superpixel Generation: Generally, pixels with the same label tend to be distributed within homogeneous regions. Recent studies have shown that superpixel segmentation techniques can effectively reveal the spectral–spatial structure of HS images. It is beneficial to localize the homogeneity of HS data for further analysis. Among various effective segmentation methods, we adopt the ERS segmentation method, which adaptively generates homogeneous regions with varying shapes and sizes based on complex spatial distribution and texture complexity. Originally designed for RGB image segmentation, ERS converts RGB images to gray scale before segmentation. Since HS consists of multiple components, a single component must be obtained before segmentation. We use the first principal component, which contains the most significant HS information. Specifically, let HS be a third-order tensor $\boldsymbol{\mathcal{X}}^1 \in \mathbb{R}^{H\times W\times B^1}$ with $H \times W$ pixels and B^1 bands, H and W representing the number of rows and columns in the data, corresponding to the pixel count along the height and width. During segmentation, we incorporate elevation information from LiDAR data, represented as a third-order tensor $\boldsymbol{\mathcal{X}}^2 \in \mathbb{R}^{H\times W\times B^2}$, with $H \times W$ pixels and B^2 bands. As shown in Fig. 2(a), we stack these data together and reshape them into a matrix $\boldsymbol{E} \in \mathbb{R}^{(B^1+B^2)\times N}$ (where $N = H \times W$). For efficient segmentation, we first perform principal component analysis (PCA) [43] on $\boldsymbol{E}$ and segment the first principal component I_p using ERS to generate M homogeneous regions (superpixels) as follows:

$$I_p = \bigcup_{i=1}^{M} \mathcal{S}_i, \quad \text{s.t. } \mathcal{S}_i \cap \mathcal{S}_j = \varnothing, \ (i \neq j) \tag{12}$$

where $\mathcal{S}_i$ is the ith local region with the same segmentation label. The constraint $\mathcal{S}_i \cap \mathcal{S}_j = \varnothing$ ensures that any two generated local regions are nonoverlapping. Furthermore, the segmentation results of $\boldsymbol{E}$ are uniformly mapped back onto both the HS data and the LiDAR data, ensuring consistent segmentation results for the HS and LiDAR images, thereby aligning the superpixel positions across different modalities.

2) Anchor Selection: For anchor-based methods, selecting a small number of anchors to reveal the distribution of all pixels is crucial. Traditional anchor selection strategies fail to choose high-quality anchors for HS images and LiDAR data because they ignore the inherent spatial distribution of pixels. To address this issue, we select informative anchors for HS and LiDAR based on superpixels generated by segmentation. These superpixel-derived anchors contain spatial information and cover all pixels in the HS and LiDAR. Compared to traditional methods, this strategy uses fewer anchors while demonstrating higher representativeness, resulting in better outcomes. Specifically, we represent the anchor matrix containing a set of anchors as $\boldsymbol{A}^v = [\boldsymbol{a}_1^v, \boldsymbol{a}_2^v, \ldots, \boldsymbol{a}_M^v] \in \mathbb{R}^{B^v\times M}$. As shown in Fig. 2(b), we determine all M anchors by averaging all pixels within the same superpixel as follows:

$$\boldsymbol{a}_j^v = \frac{1}{N_j}\sum_{i=1}^{N_j} \boldsymbol{x}_{ij}^v, \quad j = 1, 2, \ldots, M \tag{13}$$

where $\boldsymbol{x}_{ij}^v$ represents the ith pixel in the jth superpixel and N_j denotes the number of pixels in the jth superpixel. Therefore, $N = \sum_{j=1}^{M} N_j$.

In the previous step of superpixel generation, we ensured that the segmentation results of the HS image and the LiDAR

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image were consistent, with the same number of superpixels generated in corresponding pixel regions. This process guarantees that anchors are created at spatially aligned superpixel locations in both the HS and LiDAR images, ensuring that the same number of anchors is generated for superpixels at identical positions across different modalities.

C. Bipartite Graph Learning

Clustering based on bipartite graphs is a promising learning paradigm for effectively partitioning pixels into clusters. Compared to traditional graph-based clustering, which requires $\mathcal{O}(N^2)$ or $\mathcal{O}(N^3)$ time complexity, bipartite graph-based clustering is more efficient with time complexity of $\mathcal{O}(N)$. This makes it more suitable for large-scale multimodal RS data. To learn the bipartite graph $\boldsymbol{Z}^v \in \mathbb{R}^{N\times M}$ for HS and LiDAR data, as shown in Fig. 2(c), the ith pixel $\boldsymbol{x}_i^v \in \boldsymbol{X}^v$ should be connected to the jth anchor $\boldsymbol{a}_j^v \in \boldsymbol{A}^v$ with a probability Z_{ij}^v to describe their "pixel-to-anchor" neighbor relationship. Specifically, Z_{ij}^v is an element in $\boldsymbol{Z}^v$ representing the connection probability between pixel $\boldsymbol{x}_i^v$ and anchor $\boldsymbol{a}_j^v$. The larger the value of Z_{ij}^v, the smaller the distance between $\boldsymbol{x}_i^v$ and $\boldsymbol{a}_j^v$.

Therefore, bipartite graph learning of $\boldsymbol{Z}^v$ can be achieved by adaptively assigning pixels to neighboring anchors as follows:

$$\min_{\boldsymbol{Z}^v} \sum_{v=1}^{V}\sum_{i=1}^{N}\sum_{j=1}^{M} \left\|\boldsymbol{x}_i^v - \boldsymbol{a}_j^v\right\|_2^2 Z_{ij}^v \quad \text{s.t.} \boldsymbol{Z}^v\mathbf{1} = \mathbf{1}, \quad 0 \le Z_{ij}^v \le 1 \tag{14}$$

where $\|\boldsymbol{x}_i^v - \boldsymbol{a}_j^v\|_2^2$ represents the squared Euclidean distance between $\boldsymbol{x}_i^v$ and $\boldsymbol{a}_j^v$.

Notably, in both HS and LiDAR data, homogeneous regions tend to be grouped into the same superpixel. Consequently, when constructing the bipartite graph using the method described in (14), the similarity between pixels within the same superpixel and the corresponding anchor is typically higher than that with anchors of other superpixels. This indicates that these pixels are closer to their corresponding superpixel anchor. As a result, spatial information is naturally encoded into the bipartite graph $\boldsymbol{Z}^v$.

HS data often contain many redundant bands due to the high correlation between sequential feature bands, resulting in high data dimensionality and processing complexity. For instance, the HS image with hundreds of spectral channels may suffer from noise and the curse of dimensionality. To mitigate these issues, it is reasonable to introduce projection learning during the clustering process. By projecting data into a low-dimensional feature space, noise can be reduced and feature representation enhanced, improving the separability in the clustering process. Therefore, we introduce a projection matrix $\boldsymbol{W}^v \in \mathbb{R}^{B^v\times d^v}$ in the bipartite graph learning to reduce the spectral dimensionality from B^v to d^v in a new subspace as follows:

$$\min_{\boldsymbol{W}^v,\boldsymbol{Z}^v} \sum_{v=1}^{V}\sum_{i=1}^{N}\sum_{j=1}^{M} \left\|\boldsymbol{W}^{vT}\boldsymbol{x}_i^v - \boldsymbol{W}^{vT}\boldsymbol{a}_j^v\right\|_2^2 Z_{ij}^v \quad \text{s.t. } \boldsymbol{Z}^v\mathbf{1} = \mathbf{1}, \quad 0 \le Z_{ij}^v \le 1, \ \boldsymbol{W}^{vT}\boldsymbol{W}^v = \boldsymbol{I}_{d_v}. \tag{15}$$

D. Extracting Consistent Information via Low-Rank Tensor

The tensor nuclear norm (TNN) (see Definition 5) is capable of extracting consistency information among similarity graphs while filtering out potentially harmful information. Therefore, we minimize the TNN to integrate the low-rank tensor with the learning of bipartite graphs. As illustrated in Fig. 2(d), we stack each bipartite graph $\boldsymbol{Z}^v$ to form the tensor $\boldsymbol{\mathcal{Z}} \in \mathbb{R}^{N\times M\times V}$, where the vth frontal slice of the tensor corresponds to the vth bipartite graph, that is, $\boldsymbol{\mathcal{Z}}(:,:,v) = \boldsymbol{Z}^v$. Before proceeding with further detailed computations, we rotate the tensor $\boldsymbol{\mathcal{Z}}$, as depicted in Fig. 2(d), denoting the rotated tensor as $\boldsymbol{\mathcal{Z}}_r \in \mathbb{R}^{N\times V\times M}$. Upon rotation, $\boldsymbol{\mathcal{Z}}(:,:,v) = \boldsymbol{\mathcal{Z}}_r(:,v,:) = \boldsymbol{Z}^v$.

Each frontal slice of the original tensor $\boldsymbol{\mathcal{Z}}$ encodes the relationship between all pixels in a single modality and M anchors. For the vth frontal slice of the rotated tensor, $\boldsymbol{Z}_r^v \in \mathbb{R}^{N\times V}$, the ith column specifically represents the relationship between every pixel in the ith modality and the vth anchor point associated with that slice. In other words, $\boldsymbol{Z}_r^v$ already consolidates, at an individual slice level, the connections between the vth anchor point and all pixels across multiple modalities. By considering all such frontal slices together to form the rotated tensor $\boldsymbol{\mathcal{Z}}_r$, we achieve a higher level, cross-modal integration of the anchor-to-pixel relationships, as illustrated in Fig. 2(d). In addition, given that tensor singular value decomposition (t-SVD) (see Definition 4) operates on each frontal slice individually, it is intuitively apparent that, subsequent to rotation, the computational complexity significantly decreases from $\mathcal{O}(NM^2V)$ to $\mathcal{O}(NMV^2)$, where $V \ll M$. Thus, we have

$$\min_{\boldsymbol{W}^v,\boldsymbol{Z}^v} \sum_{v=1}^{V}\sum_{i=1}^{N}\sum_{j=1}^{M} \left\|\boldsymbol{W}^{vT}\boldsymbol{x}_i^v - \boldsymbol{W}^{vT}\boldsymbol{a}_j^v\right\|_2^2 Z_{ij}^v + \beta\|\boldsymbol{\mathcal{Z}}_r\|_* \quad \text{s.t. } \boldsymbol{Z}^v\mathbf{1} = \mathbf{1}, \quad 0 \le Z_{ij}^v \le 1, \ \boldsymbol{W}^{vT}\boldsymbol{W}^v = \boldsymbol{I}_{d_v}. \tag{16}$$

All in all, this approach utilizes spatial information to construct bipartite graphs, significantly reducing time complexity. It efficiently processes multimodal data through a projected clustering paradigm, which simultaneously performs clustering and DR, effectively mitigating redundant spectral interference. By stacking multimodal data into tensors, our method explores consistent low-rank structures and reduces heterogeneity-induced information divergence, thereby enhancing clustering performance.

E. Optimization

In this work, an alternating minimization strategy is employed for solving the proposed method, where we update one specific variable at a time while keeping the others fixed. The use of the ALM method significantly facilitates this process, enabling us to iteratively arrive at the optimal solution. To achieve separability of $\boldsymbol{\mathcal{Z}}_r$, we implement a variable splitting technique, introducing an auxiliary tensor variable $\boldsymbol{\mathcal{J}} \in \mathbb{R}^{N\times V\times M}$ to substitute for $\boldsymbol{\mathcal{Z}}_r$. Consequently, the proposed model can be formulated as

$$\min_{\boldsymbol{W}^v,\boldsymbol{\mathcal{J}},\boldsymbol{Z}^v} \sum_{v=1}^{V}\sum_{i=1}^{N}\sum_{j=1}^{M} \left\|\boldsymbol{W}^{vT}\boldsymbol{x}_i^v - \boldsymbol{W}^{vT}\boldsymbol{a}_j^v\right\|_2^2 Z_{ij}^v$$

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$$+\frac{\rho}{2}\left\|\mathcal{Z}_r-\mathcal{J}+\frac{\mathcal{Q}}{\rho}\right\|_F^2+\beta\|\mathcal{J}\|_*$$
$$\text{s.t. } \boldsymbol{Z}^v\mathbf{1}=\mathbf{1},\quad 0\le Z_{ij}^v\le 1,\ \boldsymbol{W}^{vT}\boldsymbol{W}^v=\boldsymbol{I}_{d_v} \tag{17}$$

where $\boldsymbol{W}^v\in\mathbb{R}^{B_v\times d_v}$, $\rho>0$ denotes the penalty factor, and $\mathcal{Q}\in\mathbb{R}^{N\times V\times M}$ represents the Lagrange multipliers.

By employing an alternating minimization strategy, problem (17) can be decomposed into three subproblems. These subproblems are then optimized iteratively, allowing for efficient variable optimization.

1) Solving $\boldsymbol{W}^v$ With Fixed $\boldsymbol{Z}^v$, $\mathcal{Q}$, and $\mathcal{J}$: In this instance, the optimization of $\boldsymbol{W}^v$ as presented in (17) evolves into

$$\min_{\boldsymbol{W}^v}\sum_{v=1}^{V}\sum_{i=1}^{N}\sum_{j=1}^{M}\left\|\boldsymbol{W}^{vT}\boldsymbol{x}_i^v-\boldsymbol{W}^{vT}\boldsymbol{a}_j^v\right\|_2^2 Z_{ij}^v$$
$$\text{s.t. } \boldsymbol{W}^{vT}\boldsymbol{W}^v=\boldsymbol{I}_{d_v}. \tag{18}$$

Note that problem (18) is independent between different v, so we can solve the following problem individually for each v:

$$\min_{\boldsymbol{W}^{vT}\boldsymbol{W}^v=\boldsymbol{I}_{d_v}}\sum_{i=1}^{N}\sum_{j=1}^{M}\left\|\boldsymbol{W}^{vT}\boldsymbol{x}_i^v-\boldsymbol{W}^{vT}\boldsymbol{a}_j^v\right\|_2^2 Z_{ij}^v. \tag{19}$$

To solve the aforementioned problem, we employ the subsequent theorem.

Theorem 1 [44]: Denote a matrix $\boldsymbol{F}\in\mathbb{R}^{N\times c}$, where the ith row is represented by $\boldsymbol{f}_i\in\mathbb{R}^{c\times 1}$, and $\boldsymbol{L}_S=\boldsymbol{D}_S-\boldsymbol{S}$ is referred to as the Laplacian matrix of $\boldsymbol{S}\in\mathbb{R}^{N\times N}$. The degree matrix $\boldsymbol{D}_S$ is defined as a diagonal matrix, with its ith diagonal element specified as $\mathbf{D}_S(i,i)=\sum_{j=1}^{N}\mathbf{S}(i,j)$. Then, we have

$$\sum_{i,j=1}^{N}\left\|\boldsymbol{f}_i-\boldsymbol{f}_j\right\|_2^2 s_{ij}=2\mathrm{tr}(\boldsymbol{F}^T\boldsymbol{L}_S\boldsymbol{F}). \tag{20}$$

Based on Theorem 1, (19) can be reformulated as

$$\begin{aligned}&\sum_{i=1}^{N}\sum_{j=1}^{M}\left\|\boldsymbol{W}^{vT}\boldsymbol{x}_i^v-\boldsymbol{W}^{vT}\boldsymbol{a}_j^v\right\|_2^2 Z_{ij}^v\\&=2\mathrm{tr}\left(\begin{bmatrix}\boldsymbol{W}^{vT}\boldsymbol{X}^v & \boldsymbol{W}^{vT}\boldsymbol{A}^v\end{bmatrix}\left(\begin{bmatrix}\boldsymbol{I}_N & \mathbf{0}\\ \mathbf{0} & \boldsymbol{D}_M^v\end{bmatrix}-\begin{bmatrix}\mathbf{0} & \boldsymbol{Z}^v\\ \boldsymbol{Z}^{vT} & \mathbf{0}\end{bmatrix}\right)\right.\\&\qquad\left.\times\begin{bmatrix}\boldsymbol{X}^T\boldsymbol{W}^v\\ \boldsymbol{A}^T\boldsymbol{W}^v\end{bmatrix}\right)\\&=2\mathrm{tr}\left(\boldsymbol{W}^{vT}\left(\boldsymbol{X}^v\boldsymbol{X}^{vT}-2\boldsymbol{X}^v\boldsymbol{Z}^v\boldsymbol{A}^{vT}+\boldsymbol{A}^v\boldsymbol{D}_M^v\boldsymbol{A}^{vT}\right)\boldsymbol{W}^v\right).\end{aligned} \tag{21}$$

Let $\boldsymbol{G}^v=\boldsymbol{X}^v\boldsymbol{X}^{vT}-2\boldsymbol{X}^v\boldsymbol{Z}^v\boldsymbol{A}^{vT}+\boldsymbol{A}^v\boldsymbol{D}_M^v\boldsymbol{A}^{vT}$, problem (19) becomes

$$\min_{\boldsymbol{W}^{vT}\boldsymbol{W}^v=\boldsymbol{I}_{d_v}}\mathrm{tr}\left(\boldsymbol{W}^{vT}\boldsymbol{G}^v\boldsymbol{W}^v\right). \tag{22}$$

We address (22) through the application of eigenvalue decomposition; the optimal solution for the variable $\boldsymbol{W}^v$ is identified as the first d^v eigenvectors, which correspond to the first d^v smallest eigenvalues of the matrix $\boldsymbol{G}^v$.

2) Solving $\mathcal{J}$ With Fixed $\boldsymbol{Z}^v$, $\mathcal{Q}$, and $\boldsymbol{W}^v$: Holding all other variables constant, $\mathcal{J}$ updates by addressing the following issue:

$$\min_{\mathcal{J}}\frac{1}{2}\left\|\mathcal{J}-\mathcal{Z}_r-\frac{\mathcal{Q}}{\rho}\right\|_F^2+\frac{\beta}{\rho}\|\mathcal{J}\|_*. \tag{23}$$

To solve the aforementioned equation, which incorporates the TNN as a regularization term, we employ the subsequent theorem.

Theorem 2 [42]: Consider a tensor $\mathcal{Y}\in\mathbb{R}^{n_1\times n_2\times n_3}$, where $l=\min(n_1,n_2)$. The tensor $\mathcal{Y}$ can be expressed in its t-SVD form as $\mathcal{Y}=\mathcal{U}*\mathcal{S}*\mathcal{V}^T$, regarding the optimization problem expressed as

$$\min_{\mathcal{X}}\frac{1}{2}\|\mathcal{X}-\mathcal{Y}\|_F^2+\tau\|\mathcal{X}\|_*. \tag{24}$$

The optimal solution for (24) can be defined as

$$\mathcal{X}^*=\Gamma_\tau(\mathcal{Y})=\mathcal{U}*\mathtt{ifft}\left(P_\tau(\tilde{\mathcal{Y}})\right)*\mathcal{V}^T \tag{25}$$

where $\tilde{\mathcal{Y}}=\mathtt{fft}(\mathcal{Y},[],3)$ denotes the tensor $\mathcal{Y}$ transformed into the frequency domain through an FFT along its third mode. The operator $P_\tau(\tilde{\mathcal{Y}})$ is defined as a tensor that facilitates the extraction of specific frequency information, while $P_\tau(\tilde{\mathcal{Y}}^k)$ represents the kth frontal slice of this tensor. This slice is expressed as $P_\tau(\tilde{\mathcal{Y}}^k)=\mathtt{diag}(\xi_1,\xi_2,\ldots,\xi_l)$, where each diagonal element ξ_i is calculated using $\xi_i=\mathtt{sign}(\sigma_i(\tilde{\mathcal{Y}}^k))$. $\sigma_i(\tilde{\mathcal{Y}}^k)$ denotes the ith singular value of the tensor slice $\tilde{\mathcal{Y}}^k$.

Based on Theorem 2, the optimal solution to (23) is presented as follows:

$$\mathcal{J}^*=\Gamma_{\frac{\beta}{\rho}}\left(\mathcal{Z}_r+\frac{\mathcal{Q}}{\rho}\right). \tag{26}$$

3) Solving $\boldsymbol{Z}^v$ With Fixed $\mathcal{Q}$, $\mathcal{J}$ and $\boldsymbol{W}^v$: The optimization with respect to $\boldsymbol{Z}^v$ as delineated in (17) becomes

$$\min_{\boldsymbol{Z}^v}\sum_{v=1}^{V}\sum_{i=1}^{N}\sum_{j=1}^{M}\left\|\boldsymbol{W}^{vT}\boldsymbol{x}_i^v-\boldsymbol{W}^{vT}\boldsymbol{a}_j^v\right\|_2^2 Z_{ij}^v+\frac{\rho}{2}\|\mathcal{Z}_r-\mathcal{T}\|_F^2$$
$$\text{s.t. } \boldsymbol{Z}^v\mathbf{1}=\mathbf{1},\quad 0\le Z_{ij}^v\le 1 \tag{27}$$

where $\mathcal{T}=\mathcal{J}-(\mathcal{Q}/\rho)$. Note that problem (27) is independent between different v, so we can solve the following problem individually for each v:

$$\min_{\boldsymbol{Z}^v}\sum_{i=1}^{N}\sum_{j=1}^{M}M_{ij}^v Z_{ij}^v+\frac{\rho}{2}\|\boldsymbol{Z}^v-\boldsymbol{T}^v\|_F^2$$
$$\text{s.t. } \boldsymbol{Z}^v\mathbf{1}=\mathbf{1},\quad 0\le Z_{ij}^v\le 1 \tag{28}$$

where $M_{ij}^v=\|\boldsymbol{W}^{v\mathrm{T}}x_i^v-\boldsymbol{W}^{v\mathrm{T}}a_j^v\|_2^2$. By simple matrix algebra, we have

$$\min_{\mathbf{Z}^v}\sum_{v=1}^{V}\langle\mathbf{M}^v-\rho\mathbf{T}^v,\mathbf{Z}^v\rangle+\frac{\rho}{2}\|\mathbf{Z}^v\|_F^2$$
$$\text{s.t. } \mathbf{Z}^v\mathbf{1}=\mathbf{1},\quad 0\le Z_{ij}\le 1. \tag{29}$$

By defining $\mathbf{R}^v=\rho\mathbf{T}^v-\mathbf{M}^v$, we can restate (29) as

$$\min_{\mathbf{Z}^v}\sum_{v=1}^{V}\left\|\mathbf{Z}^v-\frac{1}{\rho}\mathbf{R}^v\right\|_F^2$$
$$\text{s.t. } \mathbf{Z}^v\mathbf{1}=\mathbf{1},\quad 0\le\mathbf{Z}_{ij}\le 1. \tag{30}$$

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To address this optimization problem, we employ the approach proposed by [45]. In (30), each row $\boldsymbol{Z}^{v^*}(i,:)$ of $\boldsymbol{Z}^v$ within the closed-form solution $\boldsymbol{Z}^{v^*}$ is adjusted to the nearest point on $(1/\rho)\boldsymbol{R}^v(i,:)$ within the probability simplex. The probability simplex is defined as the set of all vectors whose elements sum to 1 and are nonnegative.

The entirety of the algorithm is concisely encapsulated within Algorithm 3.

Algorithm 3 Optimization Strategy for SBGC

Require: Multimodal RS data: $\boldsymbol{\mathcal{X}}^v \in \mathbb{R}^{H\times W\times B^v}$, superpixels number M, number of modalities V, cluster number c, projection dimension d^v, and tensor hyperparameters β.

1: Stack the multimodal RS data and perform PCA and ERS to generate superpixels;
2: Generate anchors $\boldsymbol{A}^v = \left[\boldsymbol{a}_1^v, \boldsymbol{a}_2^v, \cdots, \boldsymbol{a}_M^v\right] \in \mathbb{R}^{b^v\times M}$;
3: Construct the initial bipartite graph $\boldsymbol{Z}^v \in \mathbb{R}^{N\times M}$;
4: Construct multimodal tensor $\boldsymbol{\mathcal{Z}} \in \mathbb{R}^{N\times M\times V}$ and rotate;
5: Initialize $\boldsymbol{\mathcal{Q}} = \boldsymbol{\mathcal{J}} = \boldsymbol{0}$, $\rho = 10^{-5}$, $\max_\rho = 10^{10}$, $\eta = 1.1$;
6: **while** not converge **do**
7: Calculate $\{\boldsymbol{W}^v\}_{v=1}^V$ by solving Eq. (20);
8: Calculate $\boldsymbol{\mathcal{J}}$ by solving Eq. (23);
9: Calculate $\{\boldsymbol{Z}^v\}_{v=1}^V$ by solving Eq. (31);
10: Calculate $\boldsymbol{\mathcal{Q}}$ and ρ: $\boldsymbol{\mathcal{Q}} = \boldsymbol{\mathcal{Q}} + \rho(\boldsymbol{\mathcal{Z}}_r - \boldsymbol{\mathcal{J}})$, $\rho = \min(\eta\times\rho, \max_\rho)$;
11: **end while**
12: Use the K-means algorithm on $\boldsymbol{Z} = \sum_{v=1}^V \frac{\boldsymbol{Z}^v}{V}$ to determine the clustering labels;

Ensure: The final clustering result.

F. Complexity

When assessing the time complexity of SBGC, our analysis primarily focuses on the ERS segmentation algorithm and the core processes involved in solving problem (16). Specifically, we examine the time complexity associated with the ERS segmentation algorithm and the updates of the variables $\{\boldsymbol{W}^v\}_{v=1}^V$, $\boldsymbol{\mathcal{J}}$, and $\{\boldsymbol{Z}^v\}_{v=1}^V$. Their respective time complexities are $\mathcal{O}(NlogN)$, $\mathcal{O}(N\sum_{v=1}^V(\boldsymbol{B}^{v2}+\boldsymbol{B}^vM))$, $\mathcal{O}(2VMN\log(VM) + V^2MN)$, and $\mathcal{O}(VMN(c+1) + VMN\log(M))$, where c denotes the number of clusters, $\boldsymbol{B}^v$ represents the number of bands in each modality, and $V \ll \{\boldsymbol{B}^v, M\} \ll N$. Aggregating all subproblem complexities, the time complexity of SBGC is $\mathcal{O}(N\sum_{v=1}^V(\boldsymbol{B}^{v2} + \boldsymbol{B}^vM))$.

IV. Experiments

This section presents a comprehensive evaluation of the proposed SBGC algorithm. We assessed SBGC's performance by comparing it against both traditional and state-of-the-art clustering methods using three diverse HS-LiDAR datasets: Trento, MUUFL, and Houston. The comparison demonstrates SBGC's effectiveness in handling high-dimensional data and mitigating spectral variability while integrating spatial information, thus enhancing clustering accuracy and efficiency. The results underscore SBGC's ability to deliver high-quality clustering solutions for multimodal RS data, addressing the challenges associated with large-scale data and complex scenes. The comparison results are presented in Tables II–VII, with the best values for the producer's accuracy (PA) for each land cover type, overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (Kappa) highlighted in bold. A detailed introduction to the datasets and the comparison algorithms used in the experiments is provided in the following.

TABLE I
LAND-COVER TYPES OF THE TRENTO, HOUSTON, AND MUUFL GULFPORT DATASETS

Trento		MUUFL		Houston	
Class Name	#pixels	Class Name	#pixels	Class Name	#pixels
Apple trees	4034	Trees	23246	Healthy grass	1251
Buildings	2903	Mostly grass	4270	Stressed grass	1254
Ground	479	Mixed ground surface	6882	Synthetic grass	697
Wood	9123	Dirt and sand	1826	Trees	1244
Vineyard	10501	Road	6687	Soil	1242
Roads	3174	Water	466	Water	325
		Building shadow	2233	Residential	1268
		Building	6240	Commercial	1244
		Sidewalk	1385	Road	1252
		Yellow curb	183	Highway	1227
		Cloth Panels	269	Railway	1235
				Parking Lot 1	1233
				Parking Lot 2	469
				Tennis Court	428
				Running Track	660
Total	30214	Total	53687	Total	15029

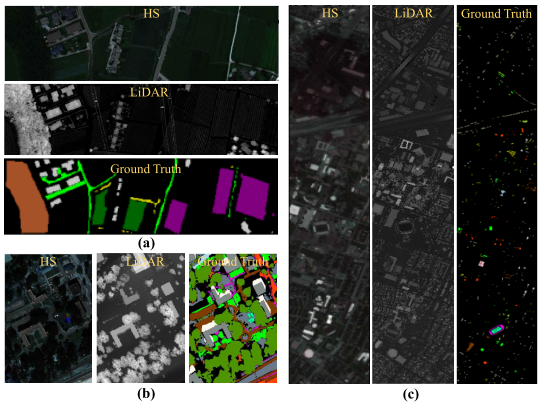


Fig. 3. False-color visualizations of HS modality, LiDAR modality, and ground truth for three datasets. Black denotes unknown classes for each ground-truth image. (a) Trento. (b) MUUFL. (c) Houston.

A. Datasets

The three HS-LiDAR datasets used in the experiments are described in detail as follows.

1) *Trento:* The Trento dataset was captured over a rural area south of the city of Trento, Italy. It consists of HS and LiDAR data with dimensions of 600 × 166 pixels and a spatial sampling of 1 m. The HS data were acquired by the AISA Eagle sensor and contain 63 spectral bands ranging from 0.40 to 0.98 μm. The LiDAR data were captured by the Optech Airborne Laser Terrain Mapper (ALTM) 3100EA sensor and contain two channels. There are six land cover categories investigated in the scene.
2) *MUUFL:* The MUUFL Gulfport scene was collected on November 8, 2010, over The University of

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TABLE II
PA, AA, OA, AND KAPPA SCORES BY OUR ALGORITHM AND COMPARISON ALGORITHMS ON THE TRENTO DATASET (THE BEST VALUE IN EACH ROW IS IN BOLD)

Class	AWMVC	FastMICE	FDAGF	FPMVS_CAG	LMVSC	SFMC	SMVSC	SBGC
Apple trees	0.4114	0.4060	0.4192	**0.5918**	0.4964	0.3461	0.5618	0.4369
Buildings	0.0006	0.4211	0.0000	0.6754	0.7598	0.6279	0.0000	**0.9655**
Ground	0.0268	0.0391	0.0235	0.0267	0.0000	0.0043	0.0340	**0.2986**
Vineyard	0.8650	0.9668	0.9483	0.7603	0.9155	0.8154	0.8394	**0.9970**
Wood	**0.9775**	0.9200	0.9027	0.9037	0.8304	0.7368	0.9405	0.9303
Roads	0.6589	0.6232	0.4952	0.5150	0.7346	0.5849	0.5154	**0.8187**
AA	0.4896	0.5824	0.5164	0.5318	0.6301	0.4886	0.5380	**0.7616**
OA	0.5793	0.6101	0.7026	0.7093	0.7412	0.6452	0.6984	**0.7816**
Kappa	0.4779	0.5196	0.6115	0.6134	0.6640	0.5326	0.6071	**0.7192**
Cpu Time	**15.5676**	138.2671	400.9348	517.1013	106.5413	73.8430	258.5940	45.9927



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 4. Clustering performance on the Trento dataset. (a) Ground-truth result. (b)–(i) Results of algorithms AWMVC, FastMICE, FDAG, FPMVS_CAG, LMVSC, SFMC, SMVSC, and SBGC, respectively.

Southern Mississippi Gulfport Campus, Long Beach, MS, USA. The HS data were acquired by the ITRES Research Ltd. (ITRES) compact airborne spectrographic imager (CASI)-1500 sensor. It has dimensions of 325 × 220 pixels with 72 spectral channels (64 available bands) ranging from 375 to 1050 nm at a spectral sampling of 10 nm. The spatial resolution of the HS data is 0.54 × 1.0 m, while that of the LiDAR-based digital surface model (DSM) is 0.60 × 0.78 m. The LiDAR data were acquired by the Gemini ALTM LiDAR sensor. There are 11 land cover categories investigated in the scene.

3) *Houston:* The Houston dataset was acquired over the area of the University of Houston, Houston, TX, USA, and neighboring areas in June 2012. It was released for the IEEE GRSS DFC 2013. The dataset consists of HS data and LiDAR-derived DSM data, both with dimensions of 349 × 1905 pixels and a spatial resolution of 2.5 m. The HS modality contains 144 spectral bands covering the spectral range from 0.38 to 1.05 μm, and the LiDAR modality contains one channel. There are 15 distinguishable land cover categories in the scene.

The details of the land cover types and the visualization of these datasets are given in Table I and Fig. 3, respectively.

B. Comparison Algorithms

1) *AWMVC (Wan et al. [46]):* AWMVC leverages matrix factorization to learn coefficient matrices from multimodal RS data, integrating them into a unified consensus matrix for optimal results.
2) *FastMICE (Huang et al. [47]):* FastMICE employs random modal groups to model cross-modal relationships, enabling a hybrid early–late fusion strategy for efficient multistage multimodal integration.
3) *FDAGF (Zhang et al. [48]):* FDAGF improves clustering by jointly optimizing the weights of multimodal multichoice anchor graphs and the significance of selections within each modality.
4) *FPMVS_CAG (Wang et al. [49]):* FPMVS_CAG integrates anchor learning and graph construction into a co-optimized framework, enhancing multimodal fusion through a unified consensus anchor graph.
5) *LMVSC (Kang et al. [50]):* LMVSC leverages multimodal RS data to construct optimal anchor graphs for each modality, combines them, and applies spectral clustering to achieve clustering results.
6) *SFMC (Nie et al. [51]):* SFMC employs feature scoring for anchor selection and integrates rank constraints in anchor graph learning, enabling direct class label prediction without postprocessing.
7) *SMVSC (Sun et al. [52]):* SMVSC unifies orthogonal consensus anchor learning and anchor graph construction, producing an optimal consensus anchor graph with improved class structure clarity.

C. Utilization of Spatial Information

The Trento dataset, which features hyperspectral data with 63 spectral bands and LiDAR data, highlights SBGC's effectiveness in leveraging spatial information. The ERS segmentation method used by SBGC constructs a bipartite graph that reduces the number of anchors while maintaining representativeness. This spatial regularization improves clustering quality by mitigating noise and spectral variability.

SBGC significantly improves the clustering accuracy for the "Buildings" and "Vineyard" classes. As shown in Table II, SBGC achieves an impressive accuracy of 0.9655 for "Buildings," which is notably higher than traditional methods such as AWMVC (0.0006) and FDAGF (0.0000). Similarly, for the "Vineyard" class, SBGC reaches an accuracy of 0.9970,

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TABLE III
PA, AA, OA, AND KAPPA SCORES BY OUR ALGORITHM AND COMPARISON ALGORITHMS ON THE MUUFL DATASET (THE BEST VALUE IN EACH ROW IS IN BOLD)

Class	AWMVC	FastMICE	FDAGF	FPMVS_CAG	LMVSC	SFMC	SMVSC	SBGC
Trees	0.9561	**0.9878**	0.9787	0.6287	0.9821	0.8807	0.6876	0.9765
Mostly grass	**0.5326**	0.1285	0.5279	0.2176	0.4864	0.3525	0.3384	0.5285
Mixed ground surface	0.6392	0.5668	0.5371	**0.8162**	0.6367	0.5044	0.7095	0.6275
Dirt and sand	0.1855	0.0878	0.2985	0.0072	0.4108	0.4318	0.0427	**0.4317**
Road	0.4992	**0.8661**	0.3485	0.3485	0.6851	0.6296	0.3468	0.8583
Water	0.0305	**0.1376**	0.0082	0.0013	0.0083	0.0012	0.0007	0.0000
Building shadow	0.1919	0.1966	0.4835	0.0098	**0.4861**	0.2690	0.1361	0.2787
Building	0.2636	0.6408	0.9098	0.9096	0.7912	0.5039	**0.9145**	0.8843
Sidewalk	0.0562	0.0037	0.0357	0.0103	0.0373	**0.1232**	0.0092	0.0140
Yellow curb	0.0005	0.0000	0.0046	**0.0635**	0.0131	0.0000	0.0137	0.0002
Cloth Panels	0.0000	0.0619	0.0227	0.0000	0.0007	0.0000	0.0000	**0.9441**
AA	0.3035	0.3935	0.4412	0.2259	0.4216	0.3433	0.2445	**0.4964**
OA	0.3500	0.3415	0.4760	0.3998	0.5044	0.3907	0.3897	**0.6165**
Kappa	0.2705	0.2550	0.4038	0.2479	0.4316	0.3140	0.2529	**0.5419**
Cpu Time	**14.3504**	16.3626	187.7522	109.8084	52.4414	33.5204	68.9494	34.0261

outperforming other methods such as FastMICE (0.9668) and FDAGF (0.9483). This performance demonstrates SBGC's strength in integrating spatial context to enhance clustering accuracy while reducing computational time, with SBGC requiring only 45.9927 s compared to the slower methods such as FDAGF (400.9348 s). With enriched spatial information, SBGC improves clustering quality by mitigating noise and spectral variability, which is also clearly reflected in the clustering result diagram, as shown in Fig. 4.

D. DR and Data Simplification

The MUUFL dataset, with hyperspectral data comprising 64 spectral channels, underscores SBGC's capability in managing high-dimensional data through its projection clustering paradigm. SBGC effectively reduces redundant spectral information while preserving significant features, simplifying data processing, and improving performance. The clustering result diagram clearly reflects the advantages of the proposed algorithm, as shown in Fig. 5.

In Table III, SBGC achieves high clustering accuracy for "Trees" (0.9765) and "Building" (0.8843), demonstrating its efficacy in handling complex hyperspectral data. This performance surpasses methods such as SFMC (0.8807 for "Trees") and FastMICE (0.6408 for "Building"), which exhibit less optimal results. SBGC's integration of DR with clustering also results in reduced processing time, requiring only 34.0261 s compared to methods such as FPMVS_CAG (109.8084 s), making it a more efficient solution.

E. Integration of Multimodal Data With Tensor-Based Approach

The Houston dataset showcases SBGC's innovative approach to integrating multimodal data. By stacking multimodal data into tensors and exploring consistent structures in the low-rank space, SBGC effectively addresses information divergence among different modalities.

As depicted in Table IV, SBGC achieves notable accuracies for "Healthy grass" (0.8606) and "Commercial" (1.0000). This performance highlights SBGC's ability to integrate rich complementary information from hyperspectral and LiDAR data, reducing information divergence and enhancing clustering stability. In contrast, methods such as LMVSC and SFMC show lower overall accuracies, with LMVSC achieving 0.4320 for "Healthy grass" and 0.5883 for "Commercial." SBGC's tensor-based approach also results in reduced processing time, with SBGC requiring 358.1551 s compared to more time-consuming methods such as FDAGF (4243.3208 s), demonstrating its efficiency in handling complex multimodal datasets. Through stacking multimodal data into tensors and exploring consistent structures in the low-rank space, SBGC improves clustering quality by mitigating noise and spectral variability, which is also clearly reflected in the clustering result diagram, as shown in Fig. 6.

F. Ablation Studies

This section delves into the experimental evaluation of the SBGC algorithm. We assess SBGC's performance using three distinct HS-LiDAR datasets: Trento, MUUFL, and Houston. The experimental results validate SBGC's three key components: DR, TNN, and ERS segmentation.

1) Impact of DR and TNN: The ablation study in Table V demonstrates the effectiveness of incorporating both DR and TNN in SBGC. The results show that using neither DR nor TNN yields the lowest clustering accuracy across all datasets. When only TNN is applied, accuracy improves, but not as much as when DR is employed alongside TNN. Specifically, on the Trento dataset, applying both DR and TNN together results in the highest accuracy (AA: 0.7616, OA: 0.7816, and Kappa: 0.7192). This suggests that DR and TNN complement each other, optimizing feature representation and data structure, thus leading to superior clustering performance.

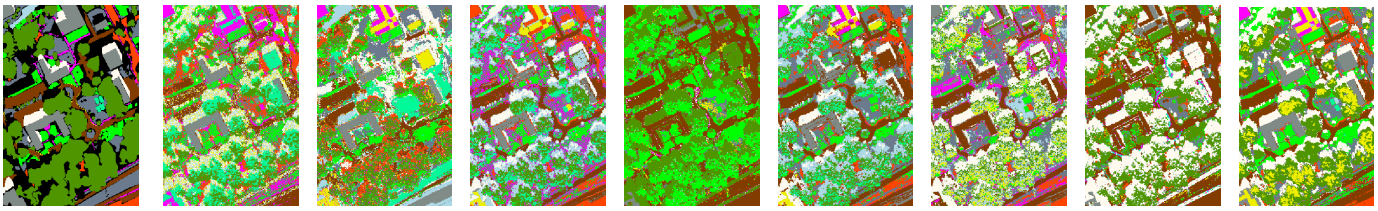
2) Effectiveness of ERS Segmentation: Table VI presents a comparison of different superpixel segmentation methods: SLIC, LSC, and ERS. The results indicate that ERS, which utilizes comprehensive background or spatial information, outperforms SLIC and LSC. For instance, on the Trento dataset, ERS achieves the highest accuracy (AA: 0.7616, OA: 0.7816,

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TABLE IV

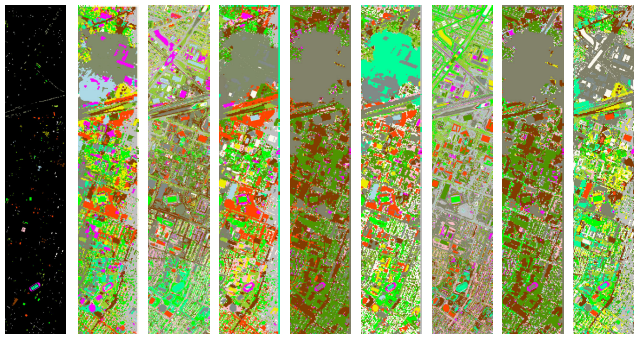
PA, AA, OA, AND KAPPA SCORES BY OUR ALGORITHM AND COMPARISON ALGORITHMS ON THE HOUSTON DATASET (THE BEST VALUE IN EACH ROW IS IN BOLD)

Class	AWMVC	FastMICE	FDAGF	FPMVS_CAG	LMVSC	SFMC	SMVSC	SBGC
Healthy grass	0.5173	0.4421	0.5062	0.0000	0.4320	0.4272	0.0000	**0.8606**
Stressed grass	0.7362	0.3876	0.7108	0.0000	0.3578	0.2473	0.0000	**0.8287**
Synthetic grass	0.4306	0.3840	**0.6402**	0.0000	0.4986	0.4050	0.0000	0.4199
Trees	0.3095	0.7363	0.0186	0.3096	0.1365	0.4449	0.3176	**0.9017**
Soil	0.4393	0.2175	0.5037	0.2754	0.5327	0.3444	0.2962	**0.7843**
Water	0.1568	0.0477	0.0309	0.0000	0.2614	0.2630	0.1224	**0.4886**
Residential	0.4768	0.4320	0.5488	0.6154	0.0000	0.3578	**0.6512**	0.5445
Commercial	0.9136	0.3858	0.9854	0.7838	0.5883	0.5447	0.1937	**1.0000**
Road	0.4678	0.5287	0.4491	0.7616	0.3365	0.1736	**0.8038**	0.0000
Highway	0.2903	0.3818	0.2864	0.0000	**0.5180**	0.1449	0.4667	0.5104
Railway	**0.4427**	0.3984	0.4091	0.1971	0.2654	0.3477	0.1825	0.3020
Parking Lot1	0.3113	0.3011	0.2557	**0.3797**	0.0472	0.1998	0.2111	0.0000
Parking Lot2	0.0192	0.0537	0.0000	0.4531	0.0000	0.0018	**0.5385**	0.0000
Tennis Court	0.0000	0.2380	0.3994	0.7212	0.0000	0.2439	0.5840	**0.5251**
Running Track	0.3825	0.0000	0.7169	0.7213	0.0000	0.0000	0.6636	**0.9255**
AA	0.4232	0.3613	0.4422	0.3000	0.3352	0.3222	0.3149	**0.5956**
OA	0.4048	0.3675	0.4458	0.2947	0.3798	0.2997	0.2970	**0.5972**
Kappa	0.3621	0.3214	0.4012	0.2351	0.3264	0.2475	0.2386	**0.5652**
CPU Time	381.6968	**209.6567**	4243.3208	4954.4023	1439.9864	698.4956	2185.6886	358.1551



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 5. Clustering performance on the MUUFL dataset. (a) Ground-truth result. (b)–(i) Results of algorithms AWMVC, FastMICE, FDAG, FPMVS_CAG, LMVSC, SFMC, SMVSC, and SBGC, respectively.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 6. Clustering performance on the Houston dataset. (a) Ground-truth result. (b)–(i) Results of algorithms AWMVC, FastMICE, FDAG, FPMVS_CAG, LMVSC, SFMC, SMVSC, and SBGC, respectively.

TABLE V

ABLATION STUDY RESULTS ON DR AND TNN (THE BEST VALUE IN EACH ROW IS IN BOLD)

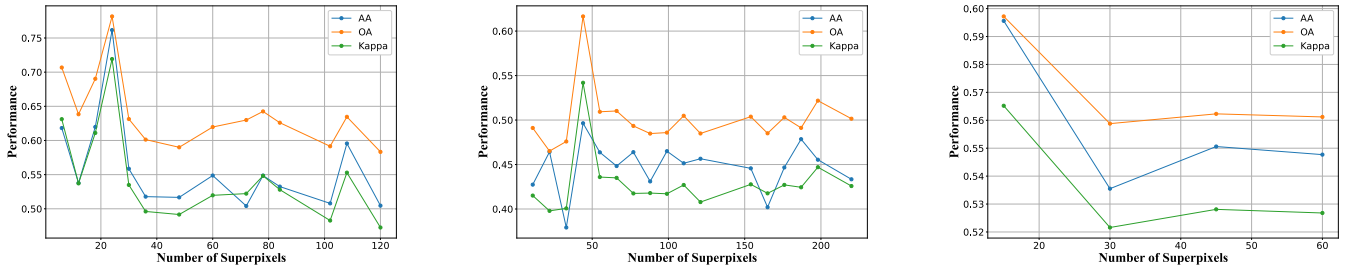
Dataset	DR	TNN	AA	OA	kappa
Trento	×	×	0.4698	0.5863	0.4744
	×	✓	0.6062	0.6624	0.5833
	✓	×	0.5934	0.6282	0.5460
	✓	✓	**0.7616**	**0.7816**	**0.7192**
MUUFL	×	×	0.4391	0.4692	0.4000
	×	✓	0.4465	0.4999	0.4237
	✓	×	0.4683	0.5073	0.4345
	✓	✓	**0.4964**	**0.6165**	**0.5419**
Houston	×	×	0.4888	0.4937	0.4512
	×	✓	0.5082	0.5686	0.5320
	✓	×	0.5555	0.5732	0.5392
	✓	✓	**0.5956**	**0.5972**	**0.5652**

and Kappa: 0.7192), while SLIC and LSC yield lower accuracies (SLIC: AA: 0.6081, OA: 0.6667, and Kappa: 0.5881; LSC: AA: 0.6061, OA: 0.6626, and Kappa: 0.5835). The key difference is that ERS incorporates the entire spatial context, unlike SLIC and LSC that rely on K-nearest neighbors for segmentation. This enhanced spatial awareness of ERS leads to better clustering precision by more accurately capturing spatial relationships within the data.

Through extensive ablation experimentation on diverse datasets, including Trento, MUUFL, and Houston, we validated the effectiveness of SBGC's key innovations.

1) *DR and TNN:* Our ablation studies demonstrated that incorporating both DR and TNN yields the highest performance metrics. This combination enhances feature representation and data analysis, leading to superior clustering accuracy.

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(a) (b) (c)

Fig. 7. Performance sensitivity to the number of superpixels for different data types. While performance initially improves with more superpixels, the relationship is not linear, and further increases can cause erratic results, indicating that optimal performance is achieved by effectively selecting representative superpixels without oversegmentation. (a) Trento. (b) MUUFL. (c) Houston.

TABLE VI
ABLATION STUDY RESULTS ON SEGMENTATION ALGORITHMS (THE BEST VALUE IN EACH ROW IS IN BOLD)

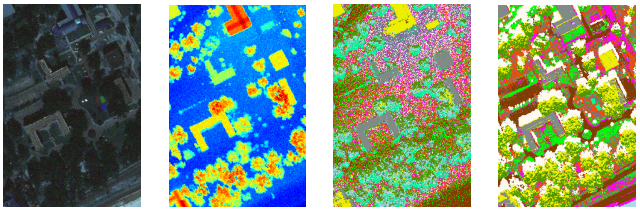
Dataset	Segmentation	AA	OA	kappa	Segment Time
Trento	SLIC	0.6081	0.6667	0.5881	0.2188
	LSC	0.6061	0.6626	0.5835	**0.2111**
	ERS	**0.7616**	**0.7816**	**0.7192**	0.3515
MUUFL	SLIC	0.4314	0.4797	0.4111	0.1641
	LSC	0.4098	0.4701	0.3960	**0.1122**
	ERS	**0.4964**	**0.6165**	**0.5419**	0.2792
Houston	SLIC	0.5668	0.5816	0.5486	**2.2709**
	LSC	0.5343	0.5452	0.5090	2.4338
	ERS	**0.5956**	**0.5972**	**0.5652**	4.1129

2) *ERS Segmentation:* By comparing different segmentation methods, we showed that ERS, which utilizes comprehensive spatial information, outperforms traditional methods such as SLIC and LSC. This superiority is attributed to ERS's ability to fully capture spatial context, thereby improving clustering precision.
3) *Performance Across Datasets:* SBGC consistently outperformed other state-of-the-art methods across various datasets. It demonstrated robust performance in integrating hyperspectral and LiDAR data, effectively handling multimodal data, and achieving better clustering accuracy and computational efficiency.

These results highlight SBGC's capability to address key challenges in multimodal RS dataset clustering, offering a robust solution that combines spatial and tensor-based methods for enhanced data analysis.

G. Number of Superpixels

Fig. 7 presents the performance variations with respect to the number of superpixels. It is evident from these figures that the number of superpixels is not linearly related to model performance, meaning that simply increasing the number of superpixels does not guarantee better results. As shown in Fig. 7, while the performance metrics such as AA, OA, and Kappa initially increase with more superpixels, there are significant fluctuations across the range of superpixel values. For instance, in Fig. 7(a), Kappa reaches a peak when the superpixel count is around 25, but after that, further increases lead to erratic performance or even a decline. This suggests that the proposed method is able to identify and select the most representative superpixels effectively without requiring an excessive number, thus maintaining both performance efficiency and accuracy. This result is crucial as it shows the model's robustness in terms of selecting optimal superpixels while avoiding oversegmentation, which can degrade performance due to the introduction of irrelevant or redundant data.



(a) (b) (c) (d)

Fig. 8. Visualization of the noise experiment. (a) Noisy HS image. (b) Noisy LiDAR image. (c) Clustering results by FastMICE (Kappa = 0.1387). (d) Clustering results by SBGC (Kappa = 0.4223).

H. Visualization Analysis

To vividly demonstrate the superior clustering performance of SBGC and its robustness to noisy data, we added the Gaussian noise with a mean of 1 and a standard deviation of 0.03 to each modality of the MUUFL HS-LiDAR multimodal RS dataset. The noisy data were then input into our proposed SBGC model and the high-performance clustering model FastMICE for comparison. Fig. 8 illustrates the visualization of the HS image and LiDAR image after the addition of Gaussian noise, as well as the clustering results produced by the FastMICE algorithm and our proposed SBGC model on the noisy multimodal RS data.

On the noise-free MUUFL dataset, FastMICE demonstrated superior clustering performance for the "Trees," "Road," and "Water" land cover types. However, its performance deteriorated significantly when applied to the noisy MUUFL dataset due to its sensitivity to noise. As shown in Fig. 8(c), the clustering results generated by FastMICE exhibit a substantial number of noisy pixels, with numerous small isolated regions and cluttered areas. The clustering boundaries are blurred, and the interiors of the regions are filled with spurious points, failing to effectively distinguish between different categories. Moreover, the presence of noise obscures fine details, making it difficult to clearly identify distinct land cover features.

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TABLE VII
AA, OA, AND KAPPA SCORES BY OUR ALGORITHM AND COMPARISON ALGORITHMS ON THE AUGSBURG DATASET (THE BEST VALUE IN EACH ROW IS IN BOLD)

Evaluation Metric	AWMVC	FastMICE	FDAGF	FPMVS_CAG	LMVSC	SFMC	SMVSC	SBGC
AA	0.3209	0.4263	0.3855	0.2562	0.3706	0.3035	0.2610	**0.4310**
OA	0.2730	0.5068	0.4142	0.4285	0.5208	0.4928	0.3864	**0.5644**
Kappa	0.1284	0.4084	0.2892	0.2066	0.3508	0.3418	0.1667	**0.4221**
CPU Time	30.8109	**30.5933**	507.8588	168.8679	111.7176	55.2763	133.4325	107.6176

In contrast, the proposed SBGC model demonstrates significantly stronger robustness to noisy data. As shown in Fig. 8(d), the clustering results obtained by SBGC are noticeably smoother, with a substantial reduction in noise points. The clustering regions are clearer and exhibit well-defined boundaries, particularly excelling in the transition areas between buildings and vegetation. In addition, the results by SBGC effectively preserve the detailed features of the geospatial objects, such as the distinct shapes of buildings and vegetation. The kappa results further underscore the substantial disparity between the two models. These differences highlight SBGC's superior noise resilience, attributed to its effective extraction of common information hidden across varying modalities by minimizing the TNN, thereby eliminating detrimental noise.

I. Extended Experiments

In this section, we emphasize the scalability of our proposed SBGC method, demonstrating its applicability to multimodal RS data with an arbitrary number of modalities. To validate this, we compared our method with baseline algorithms on a more complex dataset, Augsburb, which contains three modalities and seven ground cover types. The three heterogeneous modalities include hyperspectral (HS), SAR, and DSM. Table VII presents the comparative results.

1) Performance Comparison Across Evaluation Metrics: The comparison demonstrates that our method outperforms other algorithms in terms of AA, OA, and Kappa coefficient (Kappa). Specifically, our method achieves the highest AA of 0.4310, OA of 0.5644, and Kappa of 0.4221, surpassing all other methods. This indicates a superior ability to accurately classify and interpret the data compared to competitors.

2) Computational Efficiency Analysis: Regarding computational efficiency, our method shows a reasonable CPU time of 107.6176 s. While it is slightly slower than methods such as AWMVC and FastMICE, which have CPU times of 30.8109 and 30.5933 s, respectively, it is considerably faster than FDAGF, which has a high CPU time of 507.8588 s. The tradeoff between accuracy and computational cost is evident, as our method provides better accuracy metrics while maintaining a manageable computational burden.

In conclusion, the scalability experiments have convincingly demonstrated that the proposed SBGC model is not only well-suited for HS-LiDAR multimodal datasets but also exhibits robust adaptability to more complex and advanced multimodal datasets. This further underscores the broad applicability and significant value of SBGC in advanced RS data analysis, showcasing its potential to address complex challenges and deliver robust solutions across diverse scenarios.

V. CONCLUSION

In this work, we introduced the spatially scalable bipartite graph clustering method, addressing major challenges in hyperspectral (HS) and LiDAR data clustering, including high dimensionality, noise, and the labor required for supervised classification. Our method advances the field by fully leveraging spatial information and employing tensor-based techniques with key innovations. First, SBGC utilizes ERS segmentation to build a bipartite graph, integrating spatial context with spectral information, crucial for mitigating noise as neighboring pixels share similar spectral properties. Second, SBGC employs a projection clustering paradigm, combining DR with clustering to eliminate redundant spectral bands while retaining critical features. Third, the method's efficiency is enhanced by selecting fewer but more representative anchors, reducing computational complexity while maintaining performance. Finally, SBGC's tensor stacking explores consistent structures across different modalities, reducing heterogeneity-induced divergence and ensuring stable clustering.

Through comprehensive experiments on the Trento, MUUFL, and Houston datasets, SBGC demonstrated significant improvements in clustering accuracy, consistently outperforming state-of-the-art methods. Specifically, ablation studies revealed that combining DR with the TNN led to superior feature representation, enhancing clustering outcomes. In addition, noise experiments demonstrated SBGC's robustness to noise, while scalability experiments validated its applicability for unsupervised analysis of more advanced and complex multimodal RS data.

VI. FURTHER DISCUSSION

While SBGC demonstrates significant advancements in clustering performance, there are several areas for further exploration.

1) *Scalability and Efficiency:* Although SBGC shows competitive processing times, evaluating its scalability on larger and more complex datasets is crucial. Future work could focus on optimizing the algorithm to handle increasingly large-scale datasets efficiently without compromising performance.
2) *Adaptability to Different Modalities:* This study primarily focuses on hyperspectral and LiDAR data. Investigating SBGC's adaptability to other RS modalities, such as radar or thermal imagery, could broaden its applicability and enhance its utility in diverse scenarios.

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3) *Robustness to Noise and Outliers:* While SBGC performs well with current datasets, assessing its robustness to varying levels of noise and outliers is essential. Future research could explore integrating noise reduction techniques or developing more robust versions of SBGC to handle challenging data conditions.
4) *Real-World Applications and Validation:* Applying SBGC to real-world applications, such as environmental monitoring or urban planning, and validating its effectiveness in practical scenarios will provide valuable insights into its operational performance and potential improvements.

By addressing these areas, SBGC can be further refined and extended to achieve even greater performance and applicability in the field of RS data analysis.

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